

Total marks (84)

Attempt questions 1 – 7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 Marks) Use a SEPARATE Writing Booklet. Marks

(a) Find $\frac{d}{dx}(x \tan^{-1} 2x)$ **2**

(b) The parametric equations of a curve are given by $x = t^2$, $y = t^3 + t$. **2**
Find the Cartesian equation of the curve (that is y in terms of x).

(c) Write down the general solution of $\sin x = \frac{1}{2}$. **2**

(d) The interval AB has end points $A(5, 4)$ and $B(x, y)$. The point $P(-1, 3)$ divides AB internally in the ratio 2:3. Find the coordinates of B . **2**

(e) Evaluate $\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{4x} \right)$. **2**

(f) Use the table of standard integrals to find the exact value of **2**

$$\int_0^1 \frac{1}{\sqrt{4+x^2}} dx$$

Question 2 (12 Marks) Use a SEPARATE Writing Booklet.

Marks

- (a) Find, correct to the nearest degree, the obtuse angle between the lines $x + y - 4 = 0$ and $y = 2x + 1$.

2

- (b) Solve $\frac{2x+3}{x-4} \leq 1$.

3

- (c) Use the substitution $u = 2 - x$ to evaluate $\int_0^1 x\sqrt{2-x} \, dx$.

4

- (d) (i) Write down the domain and range of the function $y = \frac{\pi}{2} - \sin^{-1} \frac{x}{2}$.

2

- (ii) Hence sketch the function.

1

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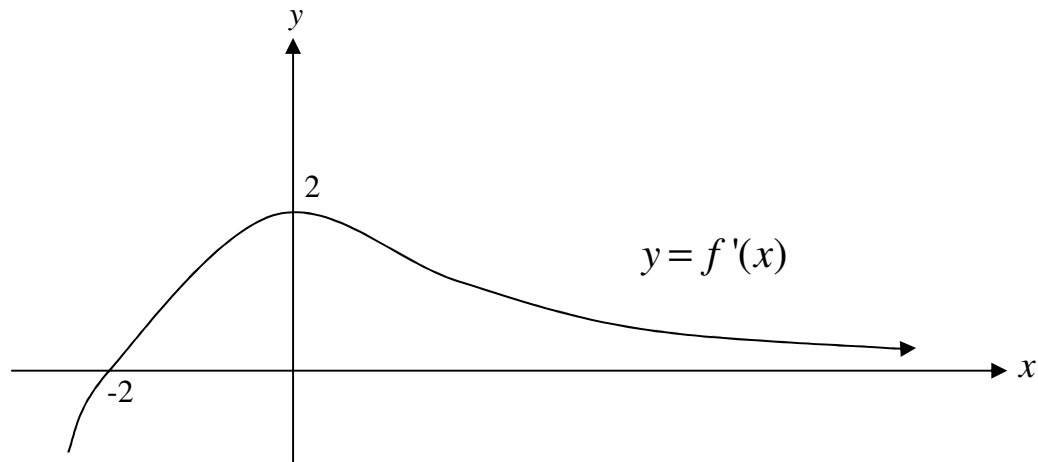
Question 3 (12 Marks) Use a SEPARATE Writing Booklet.

Marks

- (a) Find the exact value of $\cos\left(\frac{7\pi}{12}\right)$ in simplest surd form, with a rational denominator.

3

(b)



2

The diagram above shows a sketch of the gradient function of the curve $y=f(x)$.

Copy this diagram into your writing booklet.

On the same diagram, draw a possible sketch of the function $y=f(x)$, given that $f(0)=3$ and $\lim_{x \rightarrow \infty} f(x) = 6$.

- (c) Consider the point $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$.

- (i) Show that the equation of the normal to the parabola $x^2 = 4ay$ at the point P is given by $x + py = 2ap + ap^3$.

2

- (ii) Find the equation of the line which passes through the focus $S(0, a)$ and is perpendicular to the normal.

1

- (iii) If the line found in part (ii) meets the normal at N , find the coordinates of N .

2

- (iv) Show that the locus of N is a parabola and find its vertex.

2

Question 4 (12 Marks) Use a SEPARATE Writing Booklet.

Marks

(a) Determine the exact value of $\cos\left(2\sin^{-1}\left(\frac{12}{13}\right)\right)$. **2**

(b) (i) Show that the equation $x - \tan^{-1} 3x = 0$ has a root lying between $x = 1$ and $x = 2$. **1**

(ii) By taking $x = 1.5$ as an initial approximation to the root of $x - \tan^{-1} 3x = 0$, in the interval $1 < x < 2$, use one application of Newton's method to find a second approximation to this root. **2**

(c) The velocity of a particle moving in a straight line is given by **2**

$$v = 4x + 1,$$

where x is the displacement (in metres) from a fixed point O , and v is the velocity in metres per second.

Find the acceleration of the particle when it is 5 metres to the right of the origin.

(d) Newton's law of cooling states that the rate of cooling of a body is proportional to the excess of the temperature of a body above the surrounding room temperature. The temperature of a cup of chocolate drink satisfies an equation of the form $T = B + Ae^{kt}$ where T is the temperature of the drink, t is time in minutes, A and k are constants and B is the temperature of the surroundings.

The drink cools from 85°C to 70°C in three minutes in a room of temperature of 22°C .

(i) Find the value of A . **1**

(ii) Find the value of k , correct to 3 decimal places. **2**

(iii) Find the temperature of the cup of chocolate, to the nearest degree, after a further 9 minutes have passed. **2**

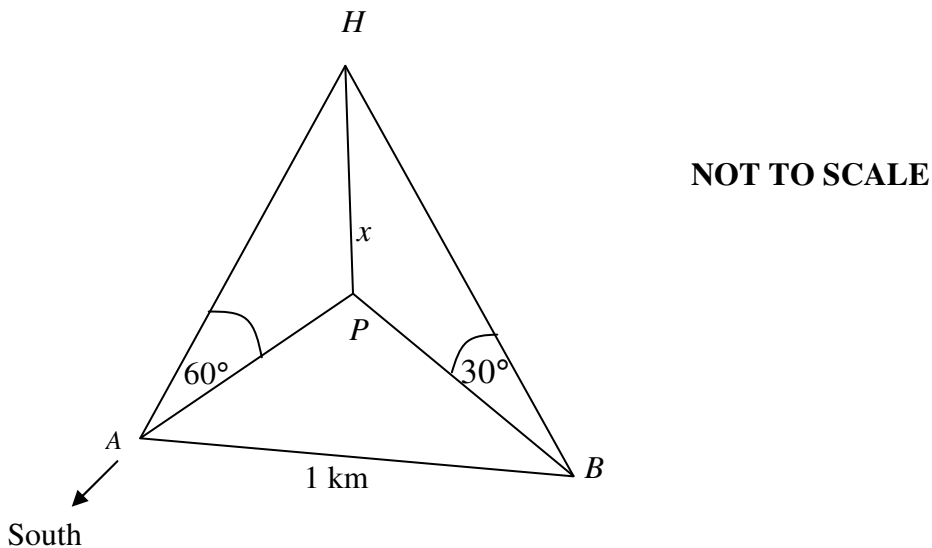
Question 5 (12 Marks) Use a SEPARATE Writing Booklet.

Marks

- (a) Suppose $\frac{\alpha}{r}$, α and αr are the real roots of the cubic equation $2x^3 - 3x^2 - 3x + 2 = 0$.

- (i) Write down the value of the sum of all three roots. **1**
- (ii) Write down the value of the product of all three roots. **1**
- (iii) Deduce that r can take on two real non-zero values and find them. **2**

- (b) Anna (A) is standing due south of Phillip (P) who is assisting an injured bush walker. A rescue helicopter (H) is hovering directly over P and lowering a stretcher. Anna measures the angle of elevation of the helicopter to be 60° from her position. Belinda (B) is 1 kilometre due east of A and measures the angle of elevation of the helicopter to be 30° . The height of the helicopter above P is x metres.



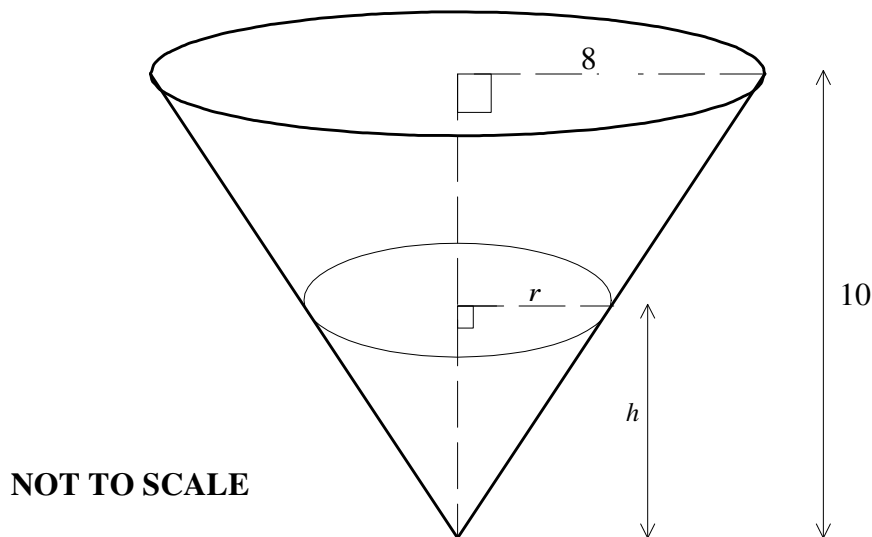
- | | | |
|------|--|----------|
| (i) | Write expressions for both AP and BP in terms of x . | 1 |
| (ii) | Hence or otherwise find the height of the helicopter correct to the nearest 10 m. | 3 |
| | | |
| (c) | Use the Principle of Mathematical Induction to show that $9^{n+2} - 4^n$ is divisible by 5 for all positive integers n . | 4 |

Question 6 (12 Marks) Use a SEPARATE Writing Booklet.

Marks

- (a) (i) State the domain and range for $f(x) = 4 - \sqrt{x-1}$. **2**
- (ii) Find the inverse function $f^{-1}(x)$ and state the domain and range for which it exists. **3**
- (iii) Sketch the graph of $f(x) = 4 - \sqrt{x-1}$ and its inverse function $f^{-1}(x)$ on the same number plane. **2**

- (b) **5**



A bulk container for emptying grain into rail trucks is in the shape of an inverted cone with base radius 8 metres and height 10 metres. The grain is released from the apex of the cone at a constant rate of $35 \text{ m}^3/\text{s}$. The depth of grain in the container at any given time is h metres and the radius of the circle formed by the top of the grain at that same time is r metres.

If the grain is released continuously until the container is empty, calculate the rate at which the radius (r) is decreasing when the depth (h) is 0.65 metres.

Question 7 (12 Marks) Use a SEPARATE Writing Booklet.

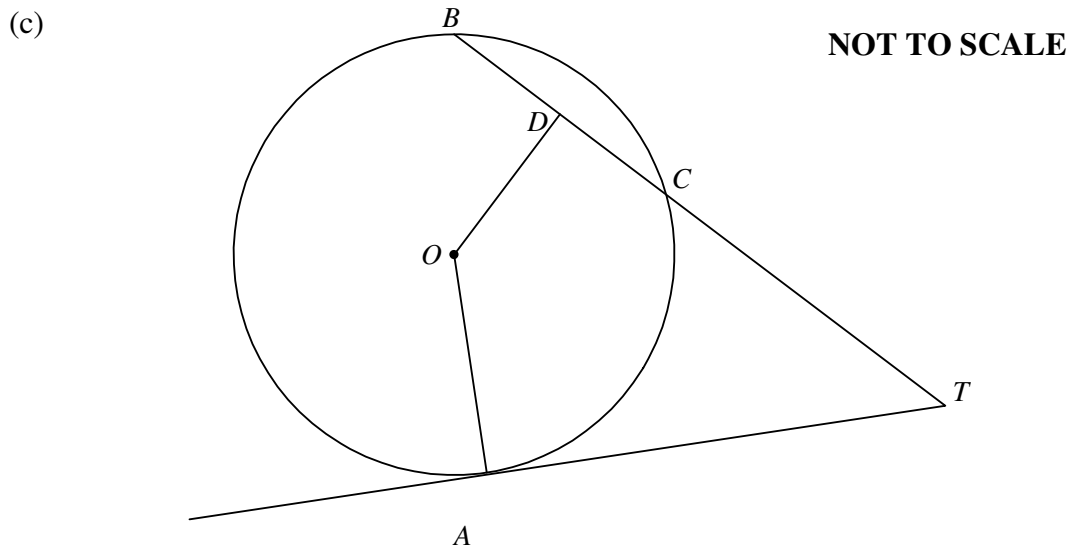
Marks

- (a) By using the t – method (that is, let $t = \tan \frac{x}{2}$) solve the equation **4**

$$\cos x + \frac{1}{\sqrt{2}} \sin x = -1,$$

for x such that $0^\circ \leq x \leq 360^\circ$

- (b) Find the volume of the solid formed by rotating about the x axis, the region bounded by $y = \cos 2x$, the x axis, from $x = 0$ to $x = \frac{\pi}{2}$. **4**



In the diagram above A , B and C are three points on a circle, centre O . The tangent at A meets BC produced at T . D is the midpoint of BC .

Copy this diagram into your writing booklet.

- (i) Prove that $AODT$ is a cyclic quadrilateral. **3**
- (ii) Explain why $\angle AOT = \angle ADT$. **1**

End of Paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$

Question 1 EXT 1 (3U) - Solutions TRIAL 2003.

$$a) \frac{d}{dx}(x \tan^{-1} 2x) = \tan^{-1} 2x + x \cdot \frac{2}{1+4x^2} \\ = \tan^{-1} 2x + \frac{2x}{1+4x^2} \quad (2)$$

$$b) x = t^2, y = t^3 + t \\ t = \pm \sqrt{x}: y = x\sqrt{x} + \sqrt{x} \text{ or } y = -x\sqrt{x} - \sqrt{x} \quad (2)$$

$$c) \sin x = \frac{1}{2} \\ x = n\pi + (-1)^n \frac{\pi}{6} \quad (2)$$

$$d) \begin{array}{cc} A(5,4) & B(x,y) \\ \frac{2x+3y}{5} = -1 & \frac{2y+3x}{5} = 3 \\ 2x+3y = -5 & 2y+3x = 15 \\ x = -10 & y = 12 \end{array} \quad (2)$$

$$e) \lim_{x \rightarrow 0} \frac{\sin 3x}{4x} = \lim_{x \rightarrow 0} \frac{3 \cos 3x}{4} = \frac{3}{4} \quad (2)$$

$$f) \int_0^1 \frac{1}{\sqrt{u+2}} du = [\ln(x + \sqrt{u+x^2})]_0^1 \\ = \ln(1 + \sqrt{5}) - \ln 2 \quad (2)$$

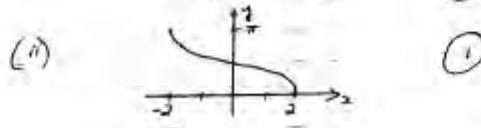
Question 2

$$(a) x+y=4 \Rightarrow 0 \quad y=2x+1 \\ m=-1 \quad m=2 \\ \tan \theta = \frac{2-(-1)}{1+2(-1)} \\ = -3 \\ \theta = 180^\circ - 72^\circ \\ = 108^\circ \quad (2)$$

$$(b) \frac{2x+3}{x-4} \leq 1 \quad x \neq 4 \\ (2x+3)(x-4) \leq (x-4)^2 \\ (x-4)[(2x+3)-(x-4)] \geq 0 \\ (x-4)(x+7) \geq 0 \\ -7 \leq x < 4 \quad (x \neq 4) \quad (2)$$

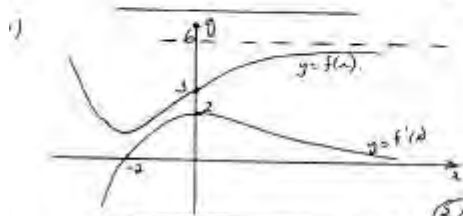
$$(c) \int_1^2 x \sqrt{2-x} dx = \int_2^1 (2-u) \sqrt{u} (-du) \quad u=2-x \\ x=2-u \quad x=0, u=2 \\ x=1, u=1 \\ = \int_2^1 (2u^{\frac{1}{2}} - u^{\frac{3}{2}}) du \\ = [2 \cdot \frac{2}{3} u^{\frac{3}{2}} - \frac{2}{5} u^{\frac{5}{2}}]_2^1 \\ = \left(\frac{4}{3} \sqrt{2} - \frac{2}{5} \sqrt{10} \right) - \left(\frac{8}{3} - \frac{2}{5} \right) \\ = 3\sqrt{2} \left(\frac{1}{3} - \frac{1}{5} \right) - \frac{14}{15} \\ = \frac{16\sqrt{2}-14}{15} \quad (4)$$

$$(d) (i) y = \frac{\pi}{2} - \sin^{-1} \frac{x}{2} \\ \text{Domain: } -1 \leq \frac{x}{2} \leq 1 \quad -2 \leq x \leq 2 \\ \text{Range: } -\frac{\pi}{2} \leq \sin^{-1} \frac{x}{2} \leq \frac{\pi}{2} \\ \frac{\pi}{2} \geq -\sin^{-1} \frac{x}{2} \geq -\frac{\pi}{2} \\ \pi \geq \frac{\pi}{2} - \sin^{-1} \frac{x}{2} \geq 0 \quad (2)$$



Question 3

$$\begin{aligned}
 1) \cos \frac{7\pi}{12} &= \cos 105^\circ = \cos (45^\circ + 60^\circ) \\
 &= \cos 45^\circ \cos 60^\circ - \sin 45^\circ \sin 60^\circ \\
 &= \frac{1}{2} \times \frac{1}{2} - \frac{1}{2} \times \frac{\sqrt{3}}{2} \\
 &= \frac{1 - \sqrt{3}}{4} \\
 &= \frac{\sqrt{3} - \sqrt{6}}{4} \quad (3)
 \end{aligned}$$

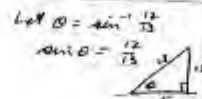


$$\begin{aligned}
 1) \quad & x^2 = 40p \\
 & \frac{dx}{dt} = \frac{20}{\sqrt{x}} \\
 \text{at } x = 20p, \frac{dx}{dt} &= \frac{2 \times 20p}{\sqrt{20p}} = p \\
 \text{Eqn of normal: } y - 20p &= -\frac{1}{p}(x - 20p) \\
 py - 20p^2 &= -x + 20p \\
 x + py &= 20p + 20p^2 \quad (A) \\
 \text{(ii) SN: } y &= px + a \\
 \text{(iii) Sub } y &= px + a \text{ into (A)} \\
 x + p(px + a) &= 20p + 20p^2 \\
 x + p^2x + pa &= 20p + 20p^2 \\
 x(1 + p^2) &= 20p(1 + p) \\
 x &= 20p, \quad y = p(20p) + a \\
 \text{N: } x &= 20p, \quad y = 20p^2 + a \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 4) \quad & p = \frac{x}{a} \therefore y = a \sqrt{\frac{x}{a}} + a \\
 & y = \frac{x}{a} + a \quad \text{rectangular parabola} \\
 & ay = x^2 + a^2 \\
 & x^2 = a(y - a) \\
 & \text{Vertex is } (0, a) \quad (3)
 \end{aligned}$$

Question 4

$$\begin{aligned}
 (a) \cos [2 \sin^{-1}(\frac{1}{3})] &= \cos 2\theta \\
 &= 1 - 2 \sin^2 \theta \\
 &= 1 - 2 \times \frac{1}{9} \\
 &= \frac{7}{9}
 \end{aligned}$$



$$\begin{aligned}
 (b) (i) \text{ Let } f(x) &= x - \tan^{-1} x \\
 f(1) &= 1 - \tan^{-1} 1 = -0.785 \\
 f(2) &= 2 - \tan^{-1} 2 = 0.579 \\
 \text{Since } f(1), f(2) &\text{ have opposite signs, } f(x) \text{ is continuous} \\
 \text{there is a root between } x &= 1, x = 2 \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad & f'(x) = 1 - \frac{1}{1+x^2} \\
 f'(1.5) &= 1 - \frac{1}{1+2.25} = 0.8588 \\
 f'(1.5) &= 1.5 - \tan^{-1} 1.5 = 0.1479 \\
 \text{Second approx'n} &= 1.5 - \frac{0.1479}{0.8588} \\
 &= 1.33 \quad (2dp) \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad & v = 4x + 1 \\
 \text{accel} &= \frac{dv}{dt} = \frac{d}{dt}(4x + 1) \\
 &= 4 \frac{dx}{dt} = 4v \\
 \text{when } x &= 5, \text{ accel} = 84 \text{ m/s}^2 \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad & T = 22 + Ae^{kt} \\
 (i) \text{ when } t &= 0, T = 85 \therefore 85 = 22 + Ae^0, A = 63 \\
 (ii) \text{ when } t &= 3, T = 70 \therefore 70 = 22 + 63e^{3k} \\
 e^{3k} &= \frac{48}{63} \\
 k &= \frac{1}{3} \ln(\frac{48}{63}) \\
 &= -0.091 \quad (3dp) \quad (2) \\
 (iii) \text{ when } t &= 12, T = 22 + 63e^{(-0.091 \times 12)} \\
 &= 43 \text{ degrees (nearest degree)} \quad (2)
 \end{aligned}$$

Question 5

$$x^2 - 3x + 2 = 0 \quad \frac{1}{x} = \frac{1}{x}, \text{ or}$$

(i) Sum of roots = 3 (1)

(ii) Product of roots = $-\frac{2}{-3} = \frac{2}{3}$ (1)

(iii) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = -1$

$$\frac{1}{x^2} = -1$$

$$x = -1$$

$$\frac{1}{-1} + (-1) + (-1) = -1$$

$$-1 - 1 - 1 = -3$$

$$-2 - 2r^2 = 5r$$

$$2r^2 + 5r + 2 = 0$$

$$(2r+1)(r+2) = 0$$

$$r = -\frac{1}{2}, -2$$

(2)

(i) $AP = x \tan 30^\circ$ $BP = x \tan 60^\circ$ (1)

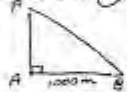
(ii) $x^2 \tan^2 60^\circ - x^2 \tan^2 30^\circ = 1000^2$

$$3x^2 - \frac{x^2}{3} = 1000^2$$

$$\frac{8x^2}{3} = 1000^2$$

$$x^2 = \frac{3 \times 1000^2}{8}$$

$$x = 610 \text{ m (nearest 10 m)} \quad (3)$$



Prove $9^{n+2} - 4^n$ is divisible by 5.

when $n=1$, $9^{1+2} - 4^1 = 9^3 - 4 = 725$

True for $n=1$.

Assume true for $n=k$, i.e. assume $9^{k+2} - 4^k = 5N$ (1)

then $n=k+1$, $9^{(k+1)+2} - 4^{k+1} = 9^{k+3} - 4^{k+1}$

$$= 9 \times 9^{k+2} - 4 \times 4^k$$

$$= 9(5N + 4^k) - 4 \times 4^k \text{ (by (1))}$$

$$= 45N + 5 \times 4^k$$

$$= 5(9N + 4^k) \text{ - divisible by 5}$$

∴ if true for $n=k$, then true for $n=k+1$.

Since true for $n=1$, then true for $n=1, 2, 3, \dots$ (4)

Question 6

(a) (i) $f(x) = 4 - \sqrt{x-1}$

Domain: $x-1 \geq 0$ i.e. $x \geq 1$

Range: $f(x) \leq 4$ (2)

(ii) $y = 4 - \sqrt{x-1}$

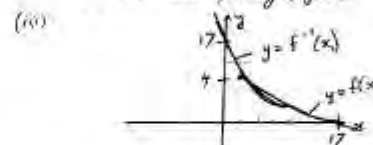
when $x = 4 - \sqrt{y-1}$

$$x-4 = -\sqrt{y-1}$$

$$(x-4)^2 = y-1$$

$$y = (x-4)^2 + 1$$

Domain: $x \leq 4$, Range: $y \geq 1$ (3)



(2)

(b) $V = \frac{1}{3} \pi r^2 h$ $\frac{r}{h} = \frac{3}{4}$

$$= \frac{1}{3} \pi \left(\frac{3}{4}h\right)^2 h$$

$$= \frac{16}{75} \pi h^3$$

$$r = \frac{3}{4}h$$

$$\frac{dr}{dt} = \frac{3}{4} \frac{dh}{dt}$$

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$35 = \frac{16\pi}{25} h^2 \cdot \frac{dr}{dt}$$

when $h = 0.65$,

$$35 = \frac{16\pi}{25} \times 0.65^2 \cdot \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{35 \times 25}{16\pi \times 0.65^2}$$

$$= 41.2$$

$$\frac{dr}{dt} = \frac{4}{5} \frac{dh}{dt}$$

$$= \frac{4}{5} \times 41.2$$

$$= 32.96$$

Radius is decreasing at 32.96 m/s . (5)

Question 5

$$x^3 - 3x^2 - 3x + 2 = 0 \quad \text{--- } x^2, x, r.$$

(i) Sum of roots = $\frac{3}{1} = 3$ (1)

(ii) Product of roots = $-\frac{2}{-1} = 2$ (1)

(iii) $x^2 + x + r = -1$

$$x^2 = -1$$

$$x = -1$$

$$\frac{-1}{1} + (-1) + (-1)r = -1$$

$$-1 - r = \frac{1}{2}$$

$$-2 - 2r^2 = 5r$$

$$2r^2 + 5r + 2 = 0$$

$$(2r+1)(r+2) = 0$$

$$r = -\frac{1}{2} \text{ or } -2$$

(2)

b) (i) $AP = x \tan 30^\circ$ $BP = x \tan 60^\circ$ (1)

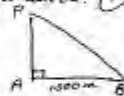
(ii) $x^2 \tan^2 60^\circ - x^2 \tan^2 30^\circ = 1000^2$

$$3x^2 - \frac{x^2}{3} = 1000000$$

$$\frac{8x^2}{3} = 1000000$$

$$x^2 = \frac{3000000}{8}$$

$$x = 610 \text{ m (nearest 10 m)}$$



c) Prove $9^{n+2} - 4^n$ is divisible by 5.

when $n=1$, $9^{1+2} - 4^1 = 9^3 - 4 = 725$

True for $n=1$

Assume true for $n=k$, i.e. $9^{k+2} - 4^k = 5N$ (1 mark)

when $n=k+1$, $9^{k+1+2} - 4^{k+1} = 9^{k+3} - 4^{k+1}$

$$= 9 \times 9^{k+2} - 4 \times 4^k$$

$$= 9(5N + 4^k) - 4 \times 4^k \text{ (by assumption)}$$

$$= 45N + 5 \times 4^k$$

$$= 5(9N + 4^k) \text{ --- divisible by 5}$$

... if true for $n=k$, then true for $n=k+1$.

Since true for $n=1$, then true for $n=1, 2, 3, \dots$

(4)

Question 6

(a) (i) $f(x) = 4 - \sqrt{x-1}$

Domain: $x-1 \geq 0$ i.e. $x \geq 1$

Range: $f(x) \leq 4$ (2)

(ii) $y = 4 - \sqrt{x-1}$

inverse: $x = 4 - \sqrt{y-1}$

$$x-4 = -\sqrt{y-1}$$

$$(x-4)^2 = y-1$$

$$y = (x-4)^2 + 1$$

Domain: $x \geq 4$, Range: $y \geq 1$ (3)



(2)

(b) $V = \frac{1}{3} \pi r^2 h$ $\frac{V}{h} = \frac{1}{3} \pi r^2$

$$= \frac{1}{3} \pi \left(\frac{r}{h}\right)^2 h$$

$$= \frac{16\pi}{75} h^3$$

$$\frac{dV}{dh} = \frac{16\pi}{25} h^2$$

$$\frac{dV}{dh} = \frac{dV}{dr} \cdot \frac{dr}{dh}$$

$$35 = \frac{16\pi}{25} h^2 \cdot \frac{dr}{dh}$$

when $h = 0.65$,

$$35 = \frac{16\pi}{25} \times 0.65^2 \cdot \frac{dr}{dh}$$

$$\frac{dr}{dh} = \frac{35 \times 25}{16\pi \times 0.65^2}$$

$$= 41.2$$

$$\frac{dr}{dt} = \frac{4}{5} \frac{dh}{dt}$$

$$= \frac{4}{5} \times 41.2$$

$$= 32.96$$

Radius is decreasing at 32.96 m/s (5)

